

10.3 THE TIME-DEPENDENT FOURIER TRANSFORM

In Section 10.2, we illustrated the use of the DFT for obtaining a frequency-domain representation of a signal composed of sinusoidal components. In that discussion, we assumed that the frequencies of the cosines did not change with time, so that no matter how long the window, the signal properties (amplitudes, frequencies, and phases) would be the same from the beginning to the end of the window. Long windows give better frequency resolution, but in practical applications of sinusoidal signal models, the signal properties (e.g., amplitude, frequency) often change with time. For example, nonstationary signal models of this type are required to describe radar, sonar, speech, and data communication signals. This conflicts with the use of long analysis windows. A single DFT estimate is not sufficient to describe such signals, and as a result, we are led to the concept of the *time-dependent Fourier transform*, also referred to as the short-time Fourier transform.¹

We define the time-dependent Fourier transform of a signal $x[n]$ as

$$X[n, \lambda] = \sum_{m=-\infty}^{\infty} x[n+m]w[m]e^{-j\lambda m}, \quad (10.18)$$

where $w[n]$ is a window sequence. In the time-dependent Fourier representation, the one-dimensional sequence $x[n]$, a function of a single discrete variable, is converted into a two-dimensional function of the time variable n , which is discrete, and the frequency variable λ , which is continuous.² Note that the time-dependent Fourier transform is periodic in λ with period 2π ; therefore, we need consider only values of λ for $0 \leq \lambda < 2\pi$ or any other interval of length 2π .

Equation (10.18) can be interpreted as the DTFT of the shifted signal $x[n+m]$, as viewed through the window $w[m]$. The window has a stationary origin, and as n changes, the signal slides past the window, so that at each value of n , a different portion of the signal is extracted by the window for Fourier analysis. As an illustration, consider the following example.

Example 10.9 Time-Dependent Fourier Transform of a Linear Chirp Signal

A continuous-time linear chirp signal is defined as

$$x_c(t) = \cos(\theta(t)) = \cos(A_0 t^2), \quad (10.19)$$

¹Further discussion of the time-dependent Fourier transform can be found in a variety of references, including Allen and Rabiner (1977), Rabiner and Schafer (1978), Crochiere and Rabiner (1983) and Quatieri (2002).

²We denote the frequency variable of the time-dependent Fourier transform by λ to maintain a distinction from the frequency variable of the conventional DTFT, which we always denote by ω . We use the mixed bracket-parenthesis notation $X[n, \lambda]$ as a reminder that n is a discrete variable, and λ is a continuous variable.

where A_0 has units of radians/s². (Such signals are called chirps because, in the auditory frequency range, short pulses sound like bird chirps.) The signal $x_c(t)$ in Eq. (10.19) is a member of the more general class of frequency modulation (FM) signals for which the *instantaneous frequency* is defined as the time derivative of the cosine argument $\theta(t)$. Therefore, in this case, the instantaneous frequency is

$$\Omega_i(t) = \frac{d\theta(t)}{dt} = \frac{d}{dt}(A_0 t^2) = 2A_0 t, \quad (10.20)$$

which varies in proportion to time; hence, the designation as a *linear* chirp signal. If we sample $x_c(t)$, we obtain the discrete-time linear chirp signal³

$$x[n] = x_c(nT) = \cos(A_0 T^2 n^2) = \cos(\alpha_0 n^2), \quad (10.21)$$

where $\alpha_0 = A_0 T^2$ has units of radians. The instantaneous frequency of the sampled chirp signal is a frequency-normalized, sampled version of the instantaneous frequency of the continuous-time signal; i.e.,

$$\omega_i[n] = \Omega_i(nT) \cdot T = 2A_0 T^2 n = 2\alpha_0 n, \quad (10.22)$$

which displays the same proportional increase with sample index n , with α_0 controlling the rate of increase. Figure 10.11 shows two 1201-sample segments of the sampled chirp signal in Eq. (10.21) with $\alpha_0 = 15\pi \times 10^{-6}$. (The samples are connected by straight lines for plotting.) Observe that over a short interval, the signal looks sinusoidal, but the spacing between peaks becomes smaller and smaller as time progresses, indicating increasing frequency with time.

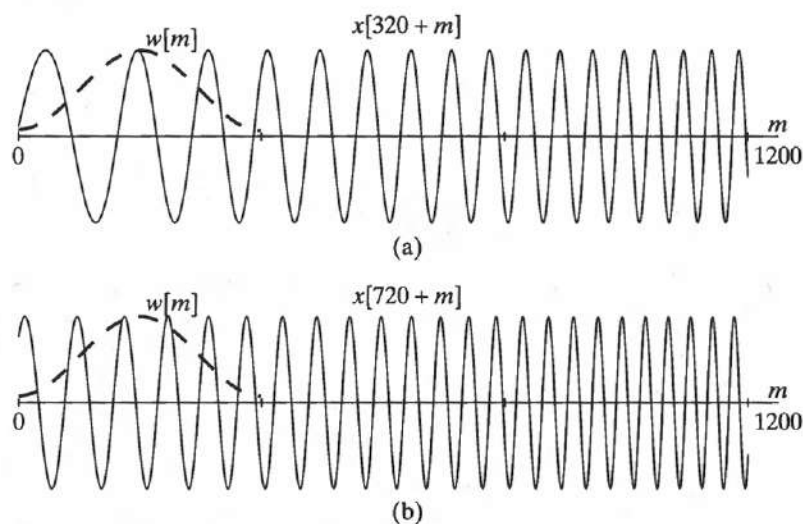


Figure 10.11 Two segments of the linear chirp signal $x[n] = \cos(\alpha_0 n^2)$ for $\alpha_0 = 15\pi \times 10^{-6}$ with a 400-sample Hamming window superimposed. (a) $X[n, \lambda]$ at $n = 320$ would be the DTFT of the top trace multiplied by the window. (b) $X[720, \lambda]$ would be the DTFT of the bottom trace multiplied by the window.

³We have seen discrete-time linear complex exponential chirp signals in Chapter 9 in the context of the chirp transform algorithm.

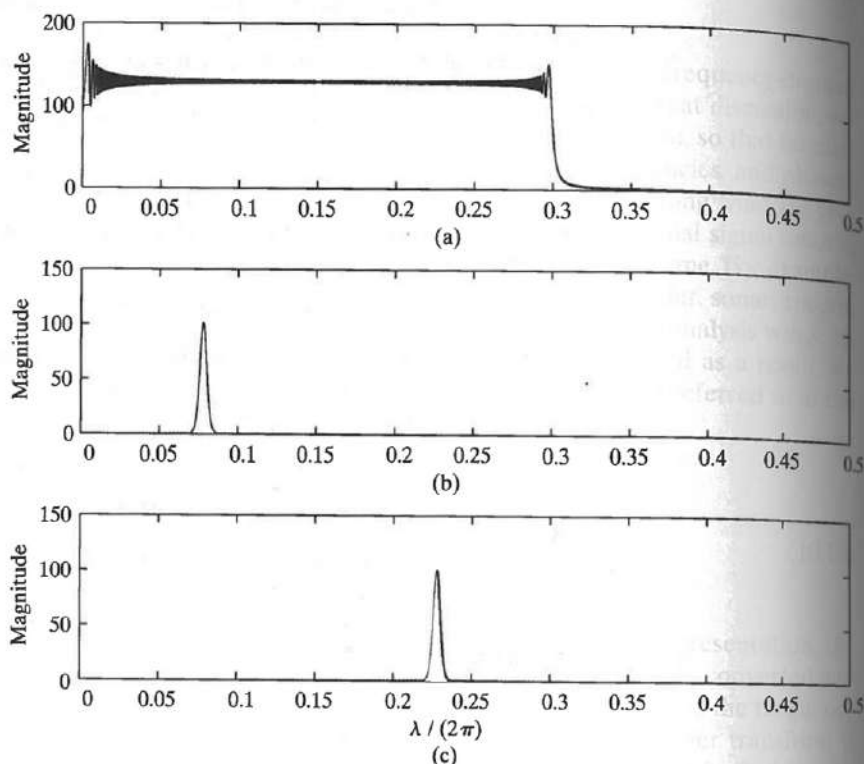


Figure 10.12 DTFTs of segments of a linear chirp signal: (a) DTFT of 20,000 samples of the signal $x[n] = \cos(\alpha_0 n^2)$. (b) DTFT of $x[5000 + m]w[m]$ where $w[m]$ is a Hamming window of length $L = 401$; i.e., $X[5000, \lambda]$. (c) DTFT of $x[15,000 + m]w[m]$ where $w[m]$ is a Hamming window of length $L = 401$; i.e., $X[15,000, \lambda]$.

The relationship of the shifted signal to the window in time-dependent Fourier analysis is also illustrated in Figure 10.11. Typically, $w[m]$ in Eq. (10.18) has finite length around $m = 0$, so that $X[n, \lambda]$ displays the frequency characteristics of the signal around time n . Figure 10.11(a) shows $x[320 + m]$ as a function of m for $0 \leq m \leq 1200$ together with a Hamming window $w[m]$ of length $L = 401$ samples. The time-dependent transform at time $n = 320$ is the DTFT of $w[m]x[320 + m]$. Similarly, Figure 10.11(b) shows the window and a later segment of the chirp signal beginning at sample $n = 720$.

Figure 10.12 illustrates the importance of the window in discrete-time Fourier analysis of time-varying signals. Figure 10.12(a) shows the DTFT of 20,000 samples (with a rectangular window) of the discrete-time chirp. Over this interval, the normalized instantaneous frequency of the chirp,

$$f_i[n] = \omega_i[n]/(2\pi) = 2\alpha_0 n/(2\pi),$$

goes from 0 to $0.00003\pi(20,000)/(2\pi) = 0.3$. This variation of the instantaneous frequency forces the DTFT representation, which involves only fixed frequencies acting over all n , to include all frequencies in that range and beyond as is evident in Figure 10.12(a). Thus, the DTFT of a long segment of the signal shows only that the

signal has a wide bandwidth in the conventional DTFT sense. On the other hand, Figures 10.12(b) and (c) show DTFTs using a 401 sample Hamming window for segments of the chirp waveform at $n = 5000$ and $15,000$, respectively. Thus, Figures 10.12(b) (c) are plots [as functions of $\lambda/(2\pi)$] of the time-dependent Fourier transform values $|X[5000, \lambda]|$ and $|X[15,000, \lambda]|$, respectively. Since the window length $L = 401$ is such that the signal does not change frequency very much across the window interval, the time-dependent Fourier transform tracks the frequency variation very well. Note that at samples 5000 and $15,000$, we would expect a peak in the time-dependent transform at $\lambda/(2\pi) = 0.00003\pi(5000)/(2\pi) = 0.075$ and $\lambda/(2\pi) = 0.00003\pi(15,000)/(2\pi) = 0.225$, respectively. This is confirmed by examination of Figures 10.12(b) and (c).

Example 10.10 Plotting $X[n, \lambda]$: The Spectrogram

In Figure 10.13, we show a display as a function of both time index n and frequency $\lambda/(2\pi)$ of the magnitude of the time-dependent Fourier transform, $|Y[n, \lambda]|$, for the signal

$$y[n] = \begin{cases} 0 & n < 0 \\ \cos(\alpha_0 n^2) & 0 \leq n \leq 20,000 \\ \cos(0.2\pi n) & 20,000 < n \leq 25,000 \\ \cos(0.2\pi n) + \cos(0.23\pi n) & 25,000 < n. \end{cases} \quad (10.23)$$

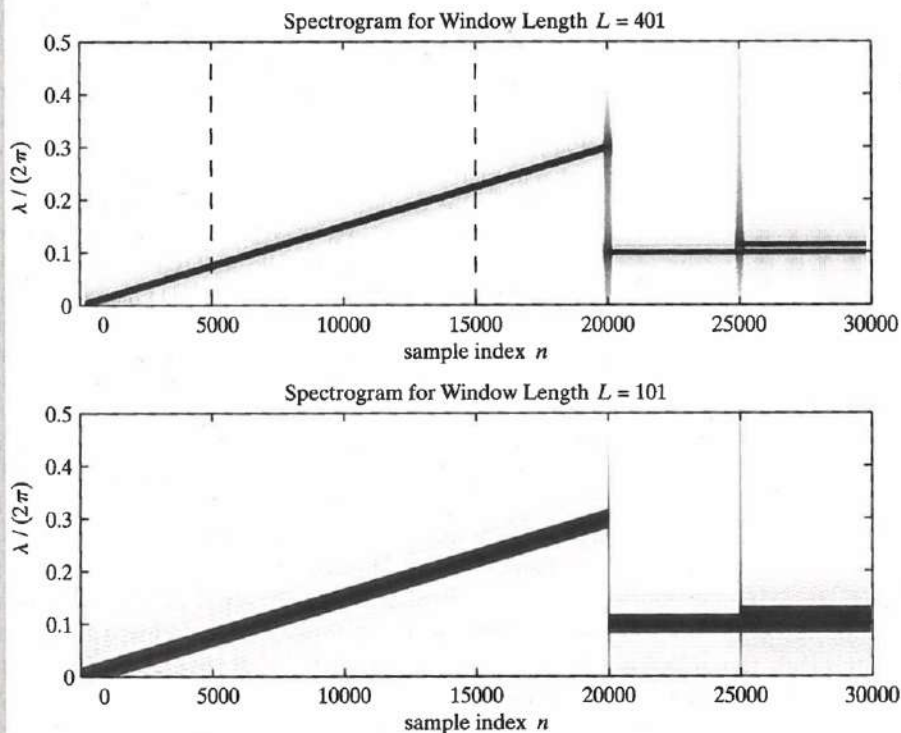


Figure 10.13 The magnitude of the time-dependent Fourier transform of $y[n]$ in Eq. (10.23): (a) Using a Hamming window of length $L = 401$. (b) Using a Hamming window of length $L = 101$.

Note that the signal $y[n]$ is equal to $x[n]$ in Eq. (10.21) in Example 10.9 for $0 \leq n \leq 20,000$, and then it abruptly changes to cosine components with fixed frequencies for $n > 20,000$. This signal was designed to make several important points about time-dependent Fourier analysis. First, consider Figure 10.13(a), which shows the time-dependent Fourier transform of $y[n]$ over the interval $0 \leq n \leq 30,000$ with a Hamming window of length $L = 401$. This display, which shows $20 \log_{10} |Y[n, \lambda]|$ as a function of $\lambda/2\pi$ in the vertical dimension, and the time index n in the horizontal dimension is called a *spectrogram*. The value $20 \log_{10} |Y[n, \lambda]|$ over a restricted range of 50 dB is represented by the darkness of the marking at (n, λ) . The plots in Figures 10.12(b) and (c) are vertical slices (shown in Figure 10.12 as magnitude) through the image at $n = 5000$ and $n = 15,000$ respectively at the locations of the dashed lines in Figure 10.13(a). Note the linear progression during the chirp interval. Also, note that during the constant-frequency intervals, the dark line remains horizontal. The width of the dark features in Figure 10.13(a) is dependent on the width of the main lobe Δ_{ml} of the DTFT of the window. Table 7.2 indicates that for the Hamming window, this width is approximately $\Delta_{ml} = 8\pi/M$ wherein $M+1$ is the window length. For a 401-point window, $\Delta_{ml}/(2\pi) = 0.01$. Thus, the two close-in-frequency cosines are clearly resolved in the interval $25,000 < n \leq 30,000$, because their normalized frequency difference is $(0.23\pi - 0.2\pi)/(2\pi) = 0.015$, which is significantly greater than the main-lobe width 0.01. Note that the vertical width of the dark sloping bar for the chirp interval is wider than the horizontal bars representing the constant-frequency intervals. This extra broadening is caused by the frequency variation across the window and is a small-scale version of the effect seen in Figure 10.12(a), wherein the variation across the 20,000-sample window is much greater.

The image in Figure 10.13(a) illustrates another important aspect of time-dependent Fourier analysis. The 401-sample window provides good frequency resolution at almost all points in time. However, note that at $n = 20,000$ and $25,000$ the signal properties change abruptly, so that for an interval of about 401 samples around these times, the window contains samples from both sides of the change. This leads to the fuzzy area wherein the signal properties are much less clearly represented by the spectrogram. We can improve the ability to resolve events in the time dimension by shortening the window. This is illustrated in Figure 10.13(b) wherein the window length is $L = 101$. The points of change are much better resolved with this window. However, the normalized main-lobe frequency width of a 101-sample Hamming window is $\Delta_{ml}/(2\pi) = 0.04$, and the two constant-frequency cosines after $n = 25,000$ are only separated by 0.015 in normalized frequency. Thus, as is clear from Figure 10.13(b), the two frequencies are not resolved with the 101-sample window, although the location of the abrupt changes in the signal are much more accurately resolved in time.

Examples 10.9 and 10.10 illustrate how the principles of discrete-time Fourier analysis that were discussed in Sections 10.1 and 10.2 can be applied to signals whose properties vary with time. Time-dependent Fourier analysis is widely used both as an analysis tool for displaying signal properties and as a representation for signals. In the latter use, it is important to develop a deeper understanding of the two-dimensional representation in Eq. (10.18).

10.3.1 Invertibility of $X[n, \lambda]$

Since $X[n, \lambda]$ is the DTFT of $x[n + m]w[m]$, the time-dependent Fourier transform is invertible if the window has at least one nonzero sample. Specifically, from the Fourier

transform synthesis equation (2.130),

$$x[n+m]w[m] = \frac{1}{2\pi} \int_0^{2\pi} X[n, \lambda] e^{j\lambda m} d\lambda, \quad -\infty < m < \infty, \quad (10.24)$$

or equivalently,

$$x[n+m] = \frac{1}{2\pi w[m]} \int_0^{2\pi} X[n, \lambda] d\lambda \quad (10.25)$$

if $w[m] \neq 0$.⁴ Thus with m chosen as any one value for which $w[m] \neq 0$, $x[n]$ for all values of n can be recovered from $X[n, \lambda]$ using Eq. (10.25).

While the above discussion shows that the time-dependent Fourier transform is an invertible transformation, Eq. (10.24) and (10.25) do not provide a computable inverse, since evaluating them requires knowing $X[n, \lambda]$ at all λ and also requires evaluating an integral. However, the inverse transform becomes a DFT when $X[n, \lambda]$ is sampled in both the time and frequency dimensions. We will discuss this matter more fully in Section 10.3.4.

10.3.2 Filter Bank Interpretation of $X[n, \lambda]$

A rearrangement of the sum in Eq. (10.18) leads to another useful interpretation of the time-dependent Fourier transform. If we make the substitution $m' = n + m$ in Eq. (10.18), then $X[n, \lambda]$ can be written as

$$X[n, \lambda] = \sum_{m'=-\infty}^{\infty} x[m']w[-(n-m')]e^{j\lambda(n-m')}. \quad (10.26)$$

Equation (10.26) can be interpreted as the convolution

$$X[n, \lambda] = x[n] * h_\lambda[n], \quad (10.27a)$$

where

$$h_\lambda[n] = w[-n]e^{j\lambda n}. \quad (10.27b)$$

From Eq. (10.27a), we see that the time-dependent Fourier transform as a function of n with λ fixed can be interpreted as the output of an LTI filter with impulse response $h_\lambda[n]$ or, equivalently, with frequency response

$$H_\lambda(e^{j\omega}) = W(e^{j(\lambda-\omega)}). \quad (10.28)$$

In general, a window that is nonzero for positive time will be called a *noncausal window*, since the computation of $X[n, \lambda]$ using Eq. (10.18) requires samples that follow sample n in the sequence. Equivalently, in the linear-filtering interpretation, the impulse response $h_\lambda[n] = w[-n]e^{j\lambda n}$ is noncausal if $w[n] = 0$ for $n < 0$. That is, a window that is nonzero for $n \geq 0$ gives a noncausal impulse response $h_\lambda[n]$ in Eq. (10.27b), whereas if the window is nonzero for $n \leq 0$, the linear filter is causal.

⁴Since $X[n, \lambda]$ is periodic in λ with period 2π , the integration in Eqs. (10.24) and (10.25) can be over any interval of length 2π .

In the definition of Eq. (10.18), the time origin of the window is held fixed, and the signal is considered to be shifted past the interval of support of the window. This effectively redefines the time origin for Fourier analysis to be at sample n of the signal. Another possibility is to shift the window as n changes, keeping the time origin for Fourier analysis fixed at the original time origin of the signal. This leads to a definition for the time-dependent Fourier transform of the form

$$\check{X}[n, \lambda] = \sum_{m=-\infty}^{\infty} x[m]w[m-n]e^{-j\lambda m}. \quad (10.29)$$

The relationship between the definitions of Eqs. (10.18) and (10.29) is easily shown to be

$$\check{X}[n, \lambda] = e^{-j\lambda n} X[n, \lambda]. \quad (10.30)$$

The definition of Eq. (10.18) is particularly convenient when we consider using the DFT to obtain samples in λ of the time-dependent Fourier transform, since, if $w[m]$ is of finite length in the range $0 \leq m \leq (L-1)$, then so is $x[n+m]w[m]$. On the other hand, the definition of Eq. (10.29) has some advantages for the interpretation of Fourier analysis in terms of filter banks. Since our primary interest is in applications of the DFT, we will base most of our discussions on Eq. (10.18).

10.3.3 The Effect of the Window

The primary purpose of the window in the time-dependent Fourier transform is to limit the extent of the sequence to be transformed, so that the spectral characteristics are approximately constant over the duration of the window. The more rapidly the signal characteristics change, the shorter the window should be. We saw in Section 10.2 that as the window becomes shorter, frequency resolution decreases. The same effect is true, of course, for $X[n, \lambda]$. On the other hand, as the window length decreases, the ability to resolve changes with time increases. Consequently, the choice of window length becomes a trade-off between frequency resolution and time resolution. This trade-off was illustrated in Example 10.10.

The effect of the window on the properties of the time-dependent Fourier transform can be seen by assuming that the signal $x[n]$ has a conventional DTFT $X(e^{j\omega})$. First, let us assume that the window is unity for all m ; i.e., assume that there is no window at all. Then, from Eq. (10.18),

$$X[n, \lambda] = X(e^{j\lambda})e^{j\lambda n}. \quad (10.31)$$

Of course, a typical window for time-dependent spectrum analysis tapers to zero, so as to select only a portion of the signal for analysis. On the other hand, as discussed in Section 10.2, the length and shape of the window are chosen so that the Fourier transform of the window is narrow in λ compared with variations in λ of the Fourier transform of the signal. Thus, the need for good time resolution and good frequency resolution often requires compromise. The Fourier transform of a typical window is illustrated in Figure 10.14(a).

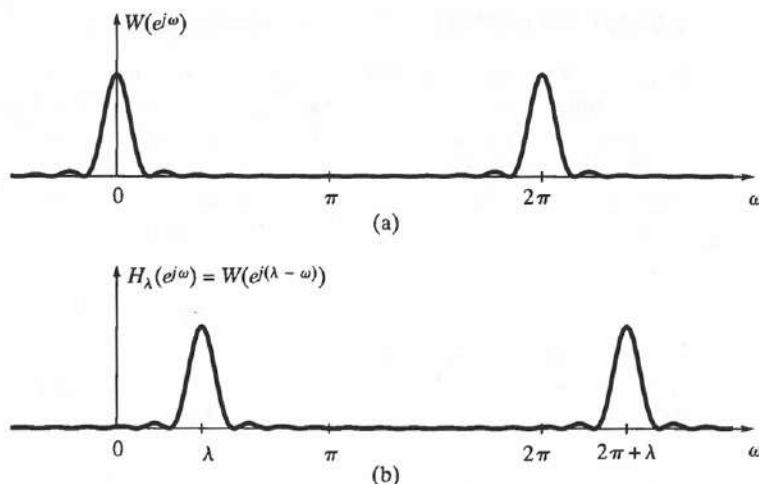


Figure 10.14 (a) Illustration of the Fourier transform of a Bartlett window for time-dependent Fourier analysis. (b) Equivalent bandpass filter for time-dependent Fourier analysis.

If we consider the time-dependent Fourier transform for fixed n , then it follows from the properties of DTFTs that

$$X[n, \lambda] = \frac{1}{2\pi} \int_0^{2\pi} e^{j\theta n} X(e^{j\theta}) W(e^{j(\lambda - \theta)}) d\theta; \quad (10.32)$$

i.e., the Fourier transform of the shifted signal is convolved with the Fourier transform of the window. This is similar to Eq. (10.2), except that in Eq. (10.2), we assumed that the signal was not successively shifted relative to the window. Here, we compute a Fourier transform for each value of n . In Section 10.2, we saw that the ability to resolve two narrowband signal components depends on the width of the main lobe of the Fourier transform of the window, whereas the degree of leakage of one component into the vicinity of the other depends on the relative side-lobe amplitude. The case of no window at all corresponds to $w[n] = 1$ for all n . In this case, $W(e^{j\omega}) = 2\pi\delta(\omega)$ for $-\pi \leq \omega \leq \pi$, which gives precise frequency resolution but no time resolution.

In the linear-filtering interpretation of Eqs. (10.27a), (10.27b), and (10.28), $W(e^{j\omega})$ typically has the lowpass characteristics depicted in Figure 10.14(a), and consequently, $H_{\lambda}(e^{j\omega})$ is a bandpass filter whose passband is centered at $\omega = \lambda$, as depicted in Figure 10.14(b). Clearly, the width of the passband of this filter is approximately equal to the width of the main lobe of the Fourier transform of the window. The degree of rejection of adjacent frequency components depends on the relative side-lobe amplitude.

The preceding discussion suggests that if we are using the time-dependent Fourier transform to obtain a time-dependent estimate of the frequency spectrum of a signal, it is desirable to taper the window to lower the side lobes and to use as long a window as feasible to improve the frequency resolution. This has already been illustrated in Examples 10.9 and 10.10, and we will consider other examples in Section 10.4. However, before doing so, we discuss the use of the DFT in explicitly evaluating the time-dependent Fourier transform.

10.3.4 Sampling in Time and Frequency

Explicit computation of $X[n, \lambda]$ can be done only on a finite set of values of λ , corresponding to sampling the time-dependent Fourier transform in the domain of its frequency variable λ . Just as finite-length signals can be exactly represented through samples of the DTFT, signals of indeterminate length can be represented through samples of the time-dependent Fourier transform, if the window in Eq. (10.18) has finite length. As an example, suppose that the window has length L with samples beginning at $m = 0$; i.e.,

$$w[m] = 0 \quad \text{outside the interval } 0 \leq m \leq L-1. \quad (10.33)$$

If we sample $X[n, \lambda]$ at N equally spaced frequencies $\lambda_k = 2\pi k/N$, with $N \geq L$, then we can recover the original windowed sequence from the sampled time-dependent Fourier transform. Specifically, if we define $X[n, k]$ to be

$$X[n, k] = X[n, 2\pi k/N] = \sum_{m=0}^{L-1} x[n+m]w[m]e^{-j(2\pi/N)km}, \quad 0 \leq k \leq N-1, \quad (10.34)$$

then $X[n, k]$ with n fixed is the DFT of the windowed sequence $x[n+m]w[m]$. Using the inverse DFT, we obtain

$$x[n+m]w[m] = \frac{1}{N} \sum_{k=0}^{N-1} X[n, k]e^{j(2\pi/N)km}, \quad 0 \leq m \leq L-1. \quad (10.35)$$

Since we assume that the window $w[m] \neq 0$ for $0 \leq m \leq L-1$, the sequence values can be recovered in the interval from n through $(n+L-1)$ using the equation

$$x[n+m] = \frac{1}{Nw[m]} \sum_{k=0}^{N-1} X[n, k]e^{j(2\pi/N)km}, \quad 0 \leq m \leq L-1. \quad (10.36)$$

The important point is that the window has finite length and that we can take at least as many samples in the λ dimension as there are nonzero samples in the window; i.e., $N \geq L$. While Eq. (10.33) corresponds to a noncausal window, we could have used a causal window with $w[m] \neq 0$ for $-(L-1) \leq m \leq 0$ or a symmetric window such that $w[m] = w[-m]$ for $|m| \leq (L-1)/2$, with L an odd integer. The use of a noncausal window in Eq. (10.34) is simply more convenient for our analysis, since it leads very naturally to the interpretation of the sampled time-dependent Fourier transform as the DFT of the windowed block of samples beginning with sample n .

Since Eq. (10.34) corresponds to sampling Eq. (10.18) in λ , it also corresponds to sampling Eqs. (10.26), (10.27a), and (10.27b) in λ . Specifically, Eq. (10.34) can be rewritten as

$$X[n, k] = x[n] * h_k[n], \quad 0 \leq k \leq N-1, \quad (10.37a)$$

where

$$h_k[n] = w[-n]e^{j(2\pi/N)kn}. \quad (10.37b)$$

Equations (10.37a) and (10.37b) can be viewed as a bank of N filters, as depicted in Figure 10.15, with the k^{th} filter having frequency response

$$H_k(e^{j\omega}) = W(e^{j[(2\pi k/N) - \omega]}). \quad (10.38)$$

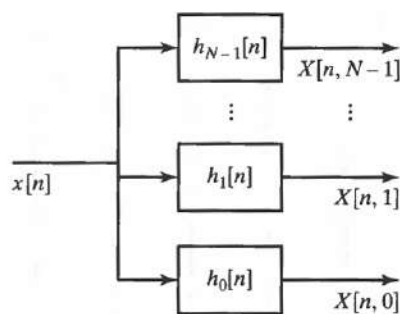


Figure 10.15 Filter bank representation of the time-dependent Fourier transform.

Our discussion suggests that $x[n]$ for $-\infty < n < \infty$ can be reconstructed if $X[n, \lambda]$ or $X[n, k]$ is sampled in the time dimension, as well. Specifically, using Eq. (10.36), we can reconstruct the signal in the interval $n_0 \leq n \leq n_0 + L - 1$ from $X[n_0, k]$, and we can reconstruct the signal in the interval $n_0 + L \leq n \leq n_0 + 2L - 1$ from $X[n_0 + L, k]$, and so on. Thus, $x[n]$ can be reconstructed exactly from the time-dependent Fourier transform sampled in both the frequency and the time dimension. In general, for the region of support of the window as specified in Eq. (10.33), we define this sampled time-dependent Fourier transform as

$$X[rR, k] = X[rR, 2\pi k/N] = \sum_{m=0}^{L-1} x[rR + m]w[m]e^{-j(2\pi/N)km}, \quad (10.39)$$

where r and k are integers such that $-\infty < r < \infty$ and $0 \leq k \leq N - 1$. To further simplify our notation, we define

$$X_r[k] = X[rR, k] = X[rR, \lambda_k), \quad -\infty < r < \infty, \quad 0 \leq k \leq N - 1, \quad (10.40)$$

where $\lambda_k = 2\pi k/N$. This notation denotes explicitly that the sampled time-dependent Fourier transform is simply a sequence of N -point DFTs of the windowed signal segments

$$x_r[m] = x[rR + m]w[m], \quad -\infty < r < \infty, \quad 0 \leq m \leq L - 1, \quad (10.41)$$

with the window position moving in jumps of R samples in time. Figure 10.16 shows lines in the $[n, \lambda]$ -plane corresponding to the region of support of $X[n, \lambda]$ and the grid of sampling points in the $[n, \lambda]$ -plane for the case $N = 10$ and $R = 3$. As we have shown, it is possible to uniquely reconstruct the original signal from such a two-dimensional discrete representation for appropriate choice of L .

Equation (10.39) involves the following integer parameters: the window length L ; the number of samples in the frequency dimension, or the DFT length N ; and the sampling interval in the time dimension, R . Although not all choices of these parameters will permit exact reconstruction of the signal, numerous combinations of N , R , and $w[n]$ and L can be used. The choice $L \leq N$ guarantees that it is possible to reconstruct the windowed segments $x_r[m]$ from the block transforms $X_r[k]$. If $R < L$, the segments overlap, but if $R > L$, some of the samples of the signal are not used and therefore cannot be reconstructed from $X_r[k]$. Thus, as one possibility, if the three sampling parameters satisfy the relation $R \leq L \leq N$, then we can (in principle) recover R samples of $x[n]$ block-by-block for all n from $X_r[k]$. Notice that each block of R samples of the

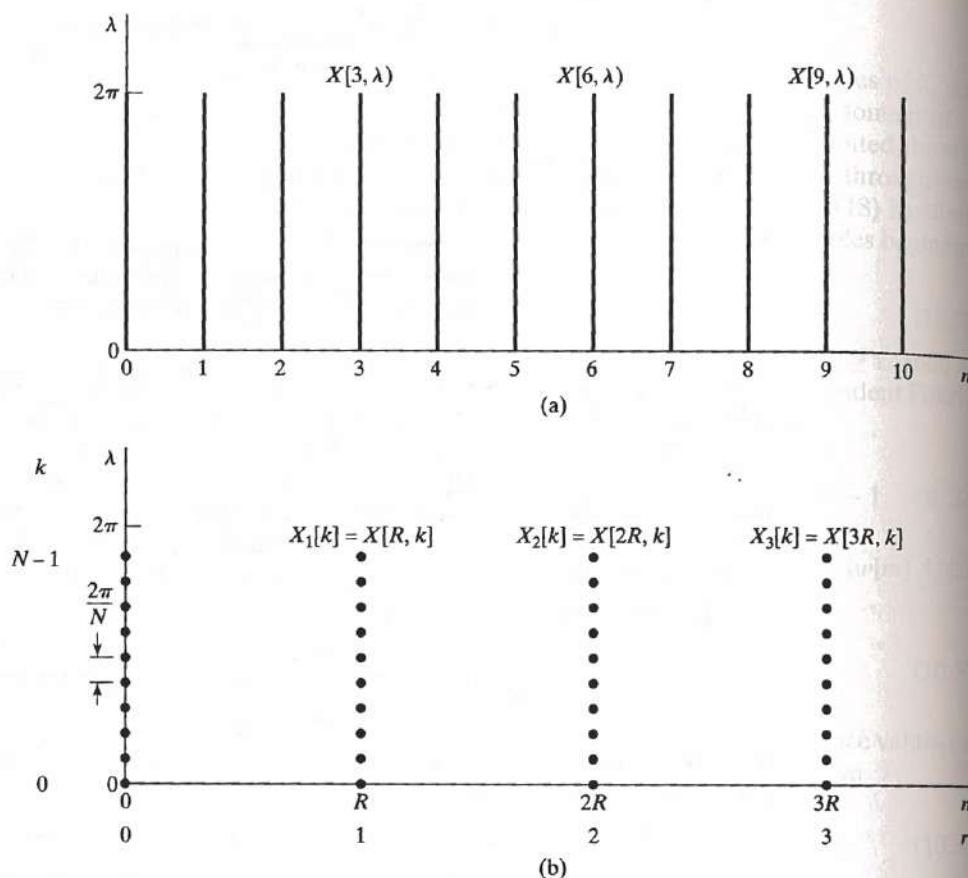


Figure 10.16 (a) Region of support for $X[n, \lambda]$. (b) Grid of sampling points in the $[n, \lambda]$ -plane for the sampled time-dependent Fourier transform with $N = 10$ and $R = 3$.

signal is represented by N complex numbers in the sampled time-dependent Fourier representation; or, if the signal is real, only N real numbers are required, due to the symmetry of the DFT.

As a specific example, the signal can be reconstructed exactly from the sampled time-dependent Fourier transform for the special case $R = L = N$. In this case, N samples of a real signal are represented by N real numbers, and this is the minimum that we could expect to achieve for an arbitrarily chosen signal. For $R = L = N$ we can recover $x_r[m] = x[rR + m]w[m]$ for $0 \leq m \leq N - 1$ by computing the inverse DFT of $X_r[k]$. Therefore, we can express $x[n]$ for $rR \leq n \leq [(r + 1)R - 1]$ in terms of the windowed segments $x_r[m]$ as

$$x[n] = \frac{x_r[n - rR]}{w[n - rR]} \quad rR \leq n \leq [(r + 1)R - 1], \quad (10.42)$$

i.e., we recover the N -point windowed segments, remove the effect of the window, and then abut the segments together to reconstruct the original sequence.

10.3.5 The Overlap-Add Method of Reconstruction

While the previous discussion verifies the possibility of theoretically exact reconstruction of the signal from its time- and frequency-sampled time-dependent Fourier transform, the demonstration proof is not a viable reconstruction algorithm when modifications are made to the time-dependent Fourier transform as is common, for example, in applications such as audio coding and noise reduction. In these applications, division by a tapering window as required in Eq. (10.42) can greatly enhance errors at the edges; therefore, the signal blocks may not fit together smoothly. In such applications, it is helpful to make R smaller than L and N so that the blocks of samples overlap. Then, if the window is properly chosen, it will not be necessary to undo the windowing as in Eq. (10.42).

Suppose that $R \leq L \leq N$. Then we can write

$$x_r[m] = x[rR + m]w[m] = \frac{1}{N} \sum_{k=0}^{N-1} X_r[k] e^{j(2\pi k/N)m} \quad 0 \leq m \leq L-1. \quad (10.43)$$

The recovered segments are shaped by the window, and their time origin is at the beginning of the window. A different approach to putting the signal back together that is more robust to changes in $X_r[k]$ is to shift the windowed segments to their original time locations rR and then simply add them together; i.e.,

$$\hat{x}[n] = \sum_{r=-\infty}^{\infty} x_r[n - rR]. \quad (10.44)$$

If we can show that $\hat{x}[n] = x[n]$ for all n , then Eqs. (10.43) and (10.44) together comprise a method for *time-dependent Fourier synthesis* having the capability of perfect reconstruction. Substituting Eq. (10.43) into Eq. (10.44) leads to the following representation of $\hat{x}[n]$:

$$\begin{aligned} \hat{x}[n] &= \sum_{r=-\infty}^{\infty} x[rR + n - rR]w[n - rR] \\ &= x[n] \sum_{r=-\infty}^{\infty} w[n - rR] \end{aligned} \quad (10.45)$$

If we define

$$\tilde{w}[n] = \sum_{r=-\infty}^{\infty} w[n - rR], \quad (10.46a)$$

then the reconstructed signal in Eq. (10.45) can be expressed as

$$\hat{x}[n] = x[n]\tilde{w}[n]. \quad (10.46b)$$

It follows from Eq. (10.46b) that the condition for perfect reconstruction is

$$\tilde{w}[n] = \sum_{r=-\infty}^{\infty} w[n - rR] = C \quad -\infty < n < \infty, \quad (10.47)$$

i.e., the shifted-by- R copies of the window must add to a constant reconstruction gain C for all n .

Note that the sequence $\tilde{w}[n]$ is a periodic sequence (with period R) comprised of time-aliased window sequences. As a simple example, consider a rectangular window $w_{\text{rect}}[n]$ of length L samples. If $R = L$, the windowed segments simply fit together block-by-block with no overlap. In this case, the condition of Eq. (10.47) is satisfied with $C = 1$, because the shifted windows fit together with no overlap and no gaps. (A simple sketch will confirm this.) If L for the rectangular window is even, and $R = L/2$ a simple analysis or sketch will again verify that the condition of Eq. (10.47) is satisfied with $C = 2$. In fact, if $L = 2^v$, the signal $x[n]$ can be perfectly reconstructed from $X_r[k]$ by the overlap-add method of Eq. (10.44) when $L \leq N$ and $R = L, L/2, \dots, 1$. The corresponding reconstruction gains would be $C = 1, 2, \dots, L$. While this demonstrates that the overlap-add method can perfectly reconstruct the original signal for some rectangular windows, and some window spacings R , the rectangular window is rarely used in time-dependent Fourier analysis/synthesis because of its poor leakage properties. Other tapered windows such as the Bartlett, Hann, Hamming, and Kaiser windows are more commonly used. Fortunately, these windows with their superior spectral isolation properties, can also produce perfect or near-perfect reconstruction from the time-dependent Fourier transform.

Two windows with which perfect reconstruction can be achieved are the Bartlett and Hann windows, which were introduced in Chapter 7 in the context of FIR filter design. They are defined again here in Eqs (10.48) and (10.49), respectively:

Bartlett (triangular)

$$w_{\text{Bart}}[n] = \begin{cases} 2n/M, & 0 \leq n \leq M/2, \\ 2 - 2n/M, & M/2 < n \leq M, \\ 0, & \text{otherwise} \end{cases} \quad (10.48)$$

Hann

$$w_{\text{Hann}}[n] = \begin{cases} 0.5 - 0.5 \cos(2\pi n/M), & 0 \leq n \leq M, \\ 0, & \text{otherwise} \end{cases} \quad (10.49)$$

As these windows are defined, the window length is $L = M + 1$ with the two end samples equal to zero.⁵ With M even and $R = M/2$, then it is easily shown for the Bartlett window that the condition of Eq. (10.47) is satisfied with $C = 1$. Figure 10.17(a) shows overlapping Bartlett windows of length $M + 1$ (first and last samples zero) when $R = M/2$. It is clear that these shifted windows add up to the reconstruction gain constant $C = 1$. Figure 10.17(b) shows the same choice of $L = M + 1$ and $R = M/2$ for the Hann window. Although it is less obvious from this plot, it is also true that these shifted windows add up for all n to the constant $C = 1$. A similar statement is also true for the Hamming window and many other windows.

⁵With these definitions, the actual number of nonzero samples is $M - 1$ for both the Bartlett and Hann windows, but the inclusion of the zero samples leads to convenient mathematical simplifications.

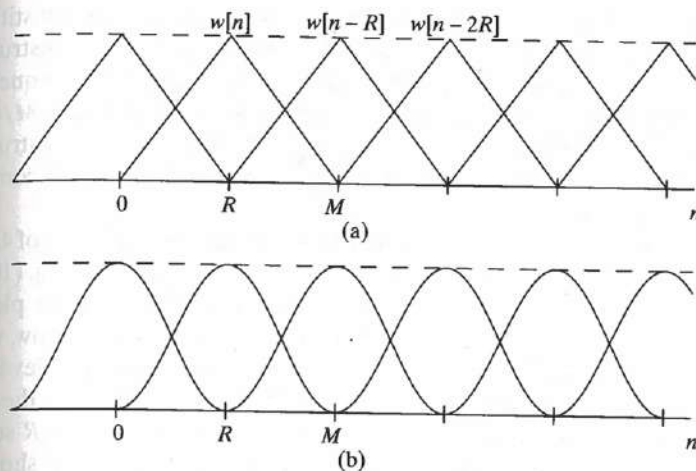


Figure 10.17 (a) Shifted $M+1$ -point Bartlett windows with $R = M/2$. (b) Shifted $M+1$ -point Hann windows with $R = M/2$. The dashed line is the periodic sequence $\tilde{w}[n]$.

Figure 7.30 gives a comparison of the DTFT of the rectangular, Bartlett and Hann windows. Note that the main-lobe width of the Bartlett and Hann windows is twice that of the rectangular window of the same length L , but the side lobes are significantly lower in amplitude for both the Bartlett and Hann windows. Thus, they and the other windows in Figure 7.30 are much preferred over the rectangular window for time-dependent Fourier analysis/synthesis.

While Figure 10.17 is intuitively plausible, it is less obvious that the Bartlett and Hann windows for $M = 2^v$ can provide perfect reconstruction for values of $R = M/2, M/4, \dots, 1$ with corresponding reconstruction gains of $M/(2R)$. To see this, it is helpful to recall that the envelope sequence $\tilde{w}[n]$ is inherently periodic with period R , so it can be represented by an inverse DFT as

$$\tilde{w}[n] = \sum_{r=-\infty}^{\infty} w[n-rR] = \frac{1}{R} \sum_{k=0}^{R-1} W(e^{j(2\pi k/R)}) e^{j(2\pi k/R)n}, \quad (10.50)$$

where $W(e^{j(2\pi k/R)})$ is the DTFT of $w[n]$ sampled at frequencies $(2\pi k/R)$, $k = 0, 1, \dots, R-1$. From Eq. (10.50) it is clear that a condition for perfect reconstruction is

$$W(e^{j(2\pi k/R)}) = 0 \quad k = 1, 2, \dots, R-1, \quad (10.51a)$$

and if Eq. (10.51a) holds, then it follows from Eq. (10.50) that the reconstruction gain is

$$C = \frac{W(e^{j0})}{R}. \quad (10.51b)$$

Problem 7.40 of Chapter 7 explores the notion that the commonly used Bartlett, Hann, Hamming, and Blackman windows can be represented in terms of rectangular windows for which it is relatively easy to obtain a closed-form expression for the DTFT of the window. In particular, Problem 7.40 gives the result that for M even, the Bartlett window defined as in Eq. (10.48) has DTFT

$$W_{\text{Bart}}(e^{j\omega}) = \left(\frac{2}{M}\right) \left(\frac{\sin(\omega M/4)}{\sin(\omega/2)}\right)^2 e^{-j\omega M/2}. \quad (10.52)$$

From Eq. (10.52) it follows that the Bartlett window Fourier transform has equally spaced zeros at frequencies $4\pi k/M$, for $k = 1, 2, \dots, M-1$. Therefore, if we choose R

so that $2\pi k/R = 4\pi k/M$ or $R = M/2$, the condition Eq. (10.51a) is satisfied. Substituting $\omega = 0$ into Eq. (10.52) gives $W_{\text{Bart}}(e^0) = M/2$, so it follows that perfect reconstruction results with $C = M/(2R) = 1$ if $R = M/2$. Choosing $R = M/2$ aligns the frequencies $2\pi k/R$ with all the zeros of $W_{\text{Bart}}(e^{j\omega})$. If M is divisible by 4, we can use $R = M/4$ and the frequencies $2\pi k/R$ will still align with zeros of $W_{\text{Bart}}(e^{j\omega})$, and the reconstruction gain will be $C = M/(2R) = 2$. If M is a power of two, R can be smaller with concomitant increase in C .

The DTFT $W_{\text{Hann}}(e^{j\omega})$ also has zeros equally spaced at integer multiples of $4\pi/M$, so exact reconstruction is also possible with the Hann window defined as in Eq. (10.49). The equally spaced zeros of $W_{\text{Bart}}(e^{j\omega})$ and $W_{\text{Hann}}(e^{j\omega})$ are evident in the plots in Figure 7.30(b) and (c), respectively. Figure 7.30(d) shows the Hamming window, which is a version of the Hann window that is optimized to minimize the side-lobe levels. As a result of the adjustment of the coefficients from 0.5 and 0.5 to 0.54 and 0.46, the zeros of $W_{\text{Hamm}}(e^{j\omega})$ are slightly displaced, so it is no longer possible to choose R so that the frequencies $2\pi k/R$ fall precisely on zeros of $W_{\text{Hamm}}(e^{j\omega})$. However, as shown in Table 7.2, the maximum side-lobe level for frequencies above $4\pi/M$ is -41 dB. Thus, the condition of Eq. (10.51a) is satisfied approximately at each of the frequencies $2\pi k/R$. Equation 10.50 shows that if Eq. (10.51a) is not satisfied exactly, $\tilde{w}[n]$ will tend to oscillate around C with period R imparting a slight amplitude modulation to the reconstructed signal.

10.3.6 Signal Processing Based on the Time-Dependent Fourier Transform

A general framework for signal processing based on the time-dependent Fourier transform is depicted in Figure 10.18. This system is based on the fact that a signal $x[n]$ can be reconstructed exactly from its time- and frequency-sampled time-dependent Fourier transform $X_r[k]$ if the window and sampling parameters are appropriately chosen, as discussed above. If the processing shown in Figure 10.18 is done so that $Y_r[k]$ maintains its integrity as a time-dependent Fourier transform, then a processed signal $y[n]$ can be reconstructed by a technique of time-dependent Fourier synthesis, such as the overlap-add method or a technique involving a bank of bandpass filters. For example, if $x[n]$ is an audio signal, $X_r[k]$ can be quantized for signal compression. The time-dependent Fourier representation provides a natural and convenient framework, wherein auditory masking phenomena can be exploited to “hide” the quantization noise. (See, for example, Bosi and Goldberg, 2003 and Spanias, Painter and Atti, 2007.) Time-dependent Fourier synthesis is then used to reconstruct a signal $y[n]$ for listening. This is the basis for MP3 audio coding, for example. Another application is audio noise suppression.

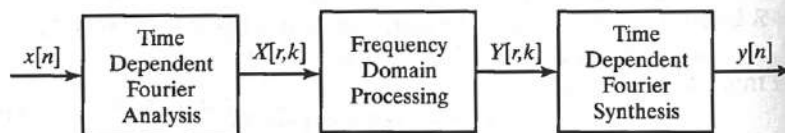


Figure 10.18 Signal processing based on time-dependent Fourier analysis/synthesis.

wherein the acoustic noise spectrum is estimated and then either subtracted from the time-dependent Fourier spectrum of the input signal or used as the basis for Wiener filtering applied to the $X_r[k]$. (See Quatieri, 2002.) These and many other applications are greatly facilitated by the FFT algorithms that are available for efficient computation of the discrete-time-dependent Fourier transform.

A discussion of applications of this type would take us too far afield; however, these kinds of block-processing techniques for discrete-time signals were also introduced in Chapter 8, when we discussed the use of the DFT for implementing the convolution of a finite-length impulse response with an input signal of indefinite length. This method of implementation of LTI systems has a useful interpretation in terms of the definitions and concepts of time-dependent Fourier analysis and synthesis, as discussed so far.

Specifically, assume that $x[n] = 0$ for $n < 0$, and suppose that we compute the time-dependent Fourier transform for $R = L$ and a rectangular window. In other words, the sampled time-dependent Fourier transform $X_r[k]$ consists of a set of N -point DFTs of segments of the input sequence

$$x_r[m] = x[rL + m], \quad 0 \leq m \leq L - 1. \quad (10.53)$$

Since each sample of the signal $x[n]$ is included, and the blocks do not overlap, it follows that

$$x[n] = \sum_{r=0}^{\infty} x_r[n - rL]. \quad (10.54)$$

Now, suppose that we define a new time-dependent Fourier transform

$$Y_r[k] = H[k]X_r[k], \quad 0 \leq k \leq N - 1, \quad (10.55)$$

where $H[k]$ is the N -point DFT of a finite-length unit sample sequence $h[n]$ such that $h[n] = 0$ for $n < 0$ and for $n > P - 1$. If we compute the inverse DFT of $Y_r[k]$, we obtain

$$y_r[m] = \frac{1}{N} \sum_{k=0}^{N-1} Y_r[k] e^{j(2\pi/N)km} = \sum_{\ell=0}^{N-1} x_r[\ell] h[(m - \ell)_N]. \quad (10.56)$$

That is, $y_r[m]$ is the N -point circular convolution of $h[m]$ and $x_r[m]$. Since $h[m]$ has length P samples and $x_r[m]$ has length L samples, it follows from the discussion of Section 8.7 that if $N \geq L + P - 1$, then $y_r[m]$ will be identical to the linear convolution of $h[m]$ with $x_r[m]$ in the interval $0 \leq m \leq L + P - 2$, and it will be zero, otherwise. Thus, it follows that if we construct an output signal

$$y[n] = \sum_{r=0}^{\infty} y_r[n - rL], \quad (10.57)$$

then $y[n]$ is the output of an LTI system with impulse response $h[n]$. The procedure just described corresponds exactly to the *overlap-add* method of block convolution. The overlap-save method discussed in Section 8.7 can also be applied within the framework of the time-dependent Fourier transform.

10.3.7 Filter Bank Interpretation of the Time-Dependent Fourier Transform

Another way to see that the time-dependent Fourier transform can be sampled in the time dimension is to recall that for fixed λ (or for fixed k if the analysis frequencies are $\lambda_k = 2\pi k/N$) the time-dependent Fourier transform is a one-dimensional sequence in time that is the output of a bandpass filter with frequency response as in Eq. (10.28).

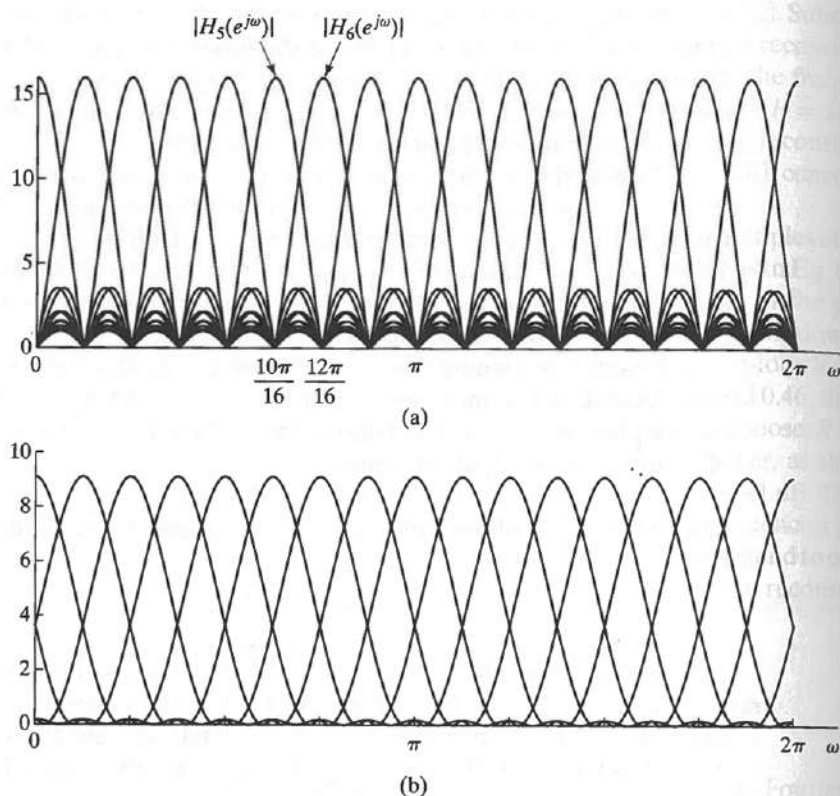


Figure 10.19 Filterbank frequency response. (a) Rectangular window. (b) Kaiser window.

This is illustrated in Figure 10.19. Figure 10.19(a) shows the equivalent set of bandpass filters corresponding to a rectangular window with $L = N = 16$. Figure 10.19 illustrates the filter bank interpretation, even for the case where L and N are much larger. When N increases, the filter bands become narrower, and the side lobes overlap with adjacent channels in the same way. Note that the passbands of the filters corresponding to the rectangular window overlap significantly, and their frequency selectivity is not good by any standard. In fact, the side lobes of any one of the bandpass filters overlap completely with several of the passbands on either side. This suggests that, in general, we might encounter a problem with aliasing in the time dimension, since the Fourier transform of any other finite-length tapering window will not be an ideal filter response either. Our discussion in Section 10.3.5, however, shows that even the rectangular window can provide perfect reconstruction with overlapping windows, in spite of the poor frequency selectivity. Although aliasing occurs in the individual bandpass filter outputs, it can be argued that the aliasing distortion cancels out when all channels are recombined in the overlap-add synthesis. This notion of alias cancellation is one of the important concepts to result from a detailed investigation of the filter bank interpretation.

If a tapering window is used, the side lobes are greatly reduced. Figure 10.19(b) shows the case for a Kaiser window of the same length as the rectangular window used

in Figure 10.19(a), i.e., $L = N = 16$. The side lobes are much smaller, but the main lobe is much wider, so the filters overlap even more. Again, the previous argument based on block processing ideas shows conclusively that we can reconstruct the original signal almost exactly from the time- and frequency-sampled time-dependent Fourier transform if R is small enough. Thus, for a Kaiser window such as in Figure 10.19(b), the sampling rate of the sequences representing each of the bandpass analysis channels could be $2\pi/R = \Delta_{\text{ml}}$, where Δ_{ml} is the width of the main lobe of the Fourier transform of the window.⁶ In the example of Figure 10.19(b), the main lobe width is approximately $\Delta_{\text{ml}} = 0.4\pi$, which implies that the time sampling interval could be $R = 5$ for nearly perfect reconstruction of the signal from $X[rR, \lambda_k]$ by the overlap-add method. More generally, in the case of the Hamming window of length $L = M + 1$ samples, for example, $\Delta_{\text{ml}} = 8\pi/M$ so nominally, the time sampling interval should be $R = M/4$. With this sampling rate in time, our discussion above shows that the signal $x[n]$ could be reconstructed nearly perfectly from $X[rR, \lambda_k]$ using a Hamming window and the overlap-add method of synthesis with $R = L/4$ and $L \leq N$.

When using the overlap-add method of analysis/synthesis, the parameters generally satisfy the relation $R \leq L \leq N$. This implies that (taking account of symmetries) the effective total number of samples (numbers) per second of the time-dependent Fourier representation $X[rR, \lambda_k]$ is a factor of N/R greater than the sample rate of $x[n]$ itself. This may not be an issue in some applications, but it presents a significant problem in data compression applications, such as audio coding. Fortunately, the filter bank point of view is the basis for showing that it is possible to choose these parameters to satisfy $R = N < L$ and still achieve nearly perfect reconstruction of the signal from its time-dependent Fourier transform. An example of such an analysis/synthesis system was discussed in Section 4.7.6, where $R = N = 2$, and the lowpass and highpass filters have impulse responses of length L , which can be as large as desired to achieve sharp cutoff filters. The two-channel filter bank can be generalized to a higher number of channels with $R = N$, and, as in the example of Section 4.7.6, polyphase techniques can be employed to increase computational efficiency. The advantage of requiring $R = N$ is that the total number of samples/s remains the same as for the input $x[n]$. As an example, Figure 10.20 shows the first few bandpass channels of the basic analysis filter bank specified by the MPEG-II audio coding standard. This filter bank performs time-dependent Fourier analysis with offset center frequencies $\lambda_k = (2k + 1)\pi/64$ using 32 real bandpass filters. Since the real bandpass filters have a pair of passbands centered at frequencies $\pm\lambda_k$, this is equivalent to 64 complex bandpass filters. In this case, the length of the impulse responses (equivalent to the window length) is $L = 513$ with the first and last samples being equal to zero. The downsampling factor is $R = 32$. Observe that the filters overlap significantly at their band edges, and downsampling by $R = 32$ causes significant aliasing distortion. However, a more detailed analysis of the complete analysis/synthesis system shows that the aliasing distortion due to the nonideal frequency responses cancels in the reconstruction process.

⁶Since, for our definition, the time-dependent Fourier transform channel signals, $X[n, \lambda_k]$, are bandpass signals centered on frequency λ_k , they can be frequency-downshifted by λ_k , so that the result is a lowpass signal in the band $\pm\Delta_{\text{ml}}$. The resulting lowpass signals have highest frequency $\Delta_{\text{ml}}/2$, so the lowest sampling rate would be $2\pi/R = \Delta_{\text{ml}}$. If $R = N$, the frequency-downshifting occurs automatically as a result of the downsampling operation.

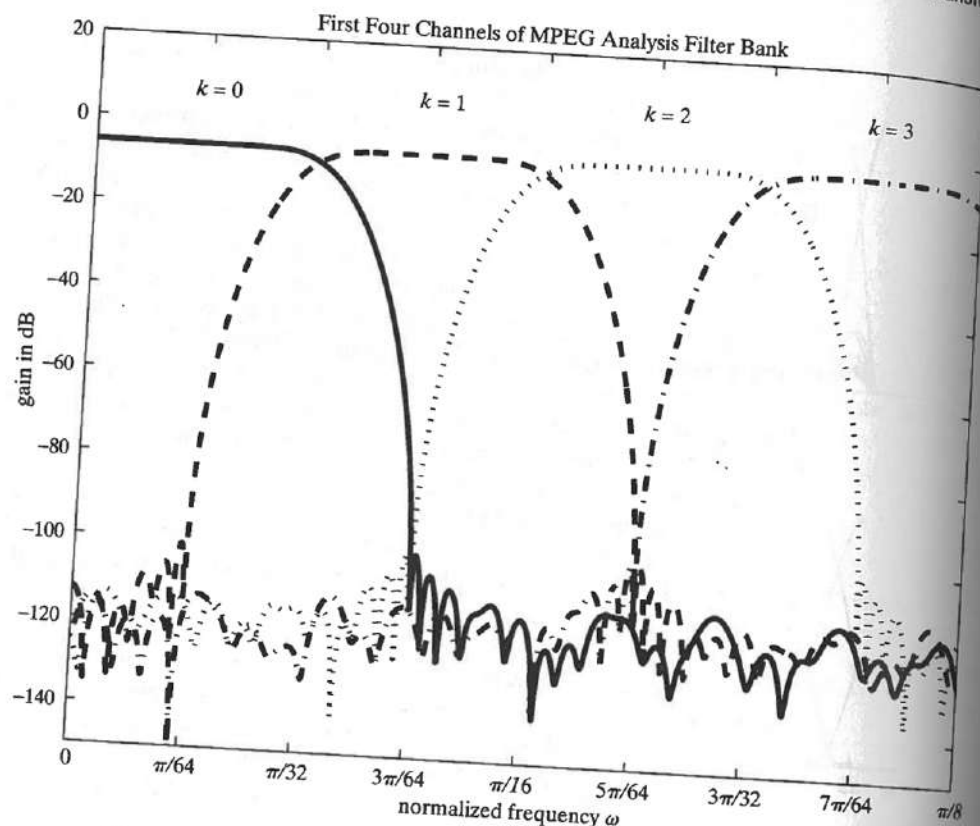


Figure 10.20 Several bandpass channels for the MPEG-II analysis filter bank.

A full-scale discussion of analysis and synthesis filter banks is beyond our scope in this chapter. An outline of such a discussion is given as the basis for Problem 10.46, and detailed discussions can be found in Rabiner and Schafer (1978), Crochiere and Rabiner (1983) and Vaidyanathan (1993).

10.4 EXAMPLES OF FOURIER ANALYSIS OF NONSTATIONARY SIGNALS

In Section 10.3.6, we considered a simple example of how the time-dependent Fourier transform can be used to implement linear filtering. In such applications, we are not so much interested in spectral resolution as in whether it is possible to reconstruct a modified signal from the modified time-dependent Fourier transform. On the other hand, the concept of the time-dependent Fourier transform is often used as a framework for a variety of techniques for obtaining spectrum estimates for nonstationary discrete-time signals, and in these applications spectral resolution, time variation, and other issues are the most important.

A nonstationary signal is a signal whose properties vary with time, for example, a sum of sinusoidal components with time-varying amplitudes, frequencies, or phases. As we will illustrate in Section 10.4.1 for speech signals and in Section 10.4.2 for Doppler

radar signals, the time-dependent Fourier transform often provides a useful description of how the signal properties change with time.

When we apply time-dependent Fourier analysis to a sampled signal, the entire discussion of Section 10.1 holds for each DFT that is computed. In other words, for each segment $x_r[n]$ of the signal, the sampled time-dependent Fourier transform $X_r[k]$ would be related to the Fourier transform of the original continuous-time signal by the processes described in Section 10.1. Furthermore, if we were to apply the time-dependent Fourier transform to sinusoidal signals with constant (i.e., nontime-varying) parameters, the discussion of Section 10.2 should also apply to each of the DFTs that we compute. When the signal frequencies do not change with time, it is tempting to assume that the time-dependent Fourier transform would vary only in the frequency dimension in the manner described in Section 10.2, but this would be true only in very special cases. For example, the time-dependent Fourier transform will be constant in the time dimension if the signal is periodic with period N_p and $L = \ell_0 N_p$ and $R = r_0 N_p$, where ℓ_0 and r_0 are integers; i.e., the window includes exactly ℓ_0 periods, and the window is moved by exactly r_0 periods between computations of the DFT. In general, even if the signal is exactly periodic, the varying phase relationships that would result as different segments of the waveform are shifted into the analysis window would cause the time-dependent Fourier transform to vary in the time dimension. However, for stationary signals, if we use a window that tapers to zero at its ends, the magnitude $|X_r[k]|$ will vary only slightly from segment to segment, with most of the variation of the complex time-dependent Fourier transform occurring in the phase.

10.4.1 Time-Dependent Fourier Analysis of Speech Signals

Speech is produced by excitation of an acoustic tube, the *vocal tract*, which is terminated on one end by the lips and on the other end by the glottis. There are three basic classes of speech sounds:

- *Voiced sounds* are produced by exciting the vocal tract with quasi-periodic pulses of airflow caused by the opening and closing of the glottis.
- *Fricative sounds* are produced by forming a constriction somewhere in the vocal tract and forcing air through the constriction so that turbulence is created, thereby producing a noise-like excitation.
- *Plosive sounds* are produced by completely closing off the vocal tract, building up pressure behind the closure, and then abruptly releasing the pressure.

Detailed discussions of models for the speech signal and applications of the time-dependent Fourier transform are found in texts such as Flanagan (1972), Rabiner and Schafer (1978), O'Shaughnessy (1999), Parsons (1986) and Quatieri (2002).

With a constant vocal tract shape, speech can be modeled as the response of an LTI system (the vocal tract) to a quasiperiodic pulse train for voiced sounds or wideband noise for unvoiced sounds. The vocal tract is an acoustic transmission system characterized by its natural frequencies, called *formants*, which correspond to resonances in its frequency response. In normal speech, the vocal tract changes shape relatively slowly with time as the tongue and lips perform the gestures of speech, and thus it can be

modeled as a slowly time-varying filter that imposes its frequency-response properties on the spectrum of the excitation. A typical speech waveform is shown in Figure 10.21.

From this brief description of the process of speech production and from Figure 10.21, we see that speech is definitely a nonstationary signal. However, as illustrated

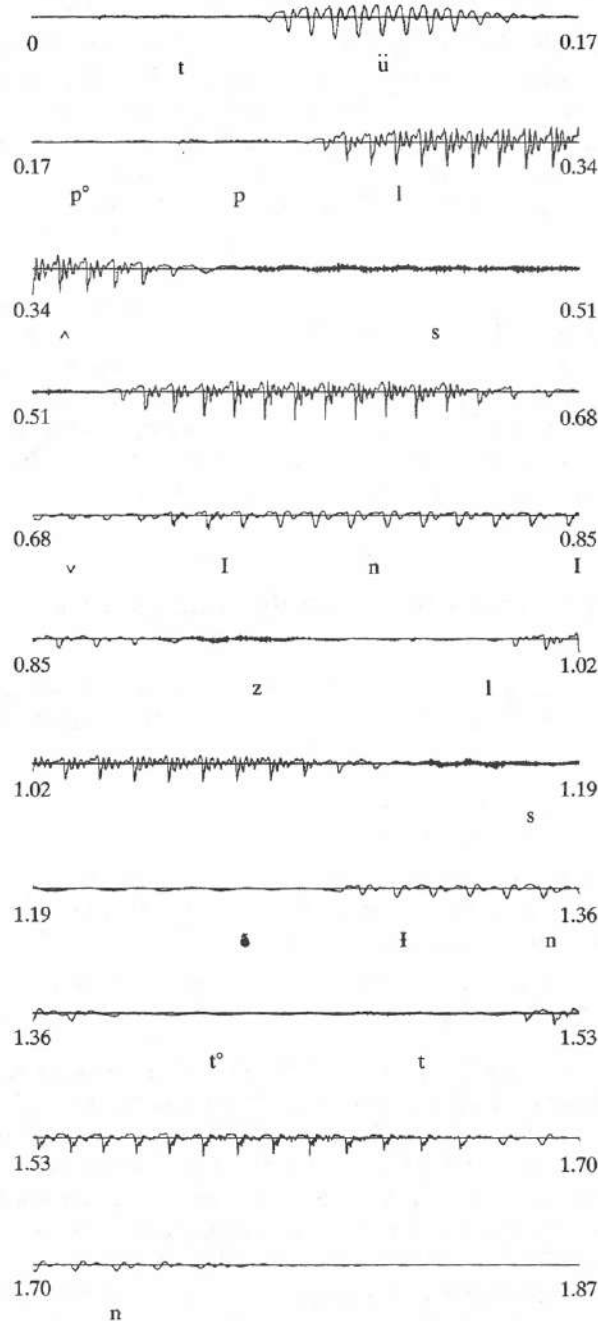


Figure 10.21 Waveform of the speech utterance "Two plus seven is less than ten." Each line is 0.17 s in duration. The time-aligned phonemic transcript is indicated below the waveform. The sampling rate is 16,000 samples/s, so each line represents 2720 samples.

in the figure, the characteristics of the signal can be assumed to remain essentially constant over time intervals on the order of 30 or 40 ms. The frequency content of the speech signal may range up to 15 kHz or higher, but speech is highly intelligible even when bandlimited to frequencies below about 3 kHz. Commercial telephone systems, for example, typically limit the highest transmitted frequency to about 3 kHz. A standard sampling rate for digital telephone communication systems is 8000 samples/s.

Figure 10.21 shows that the waveform consists of a sequence of quasiperiodic *voiced* segments interspersed with noise-like *unvoiced* segments. This figure suggests that if the window length L is not too long, the properties of the signal will not change appreciably from the beginning of the segment to the end. Thus, the DFT of a windowed speech segment should display the frequency-domain properties of the signal at the time corresponding to the window location. For example, if the window length is long enough so that the fundamental frequency and its harmonics are resolved, the DFT of a windowed segment of voiced speech should show a series of peaks at integer multiples of the fundamental frequency of the signal in that interval. This would normally require that the window span several periods of the waveform. If the window is too short, then the harmonics will not be resolved, but the general spectrum shape will still be evident. This is typical of the trade-off between frequency resolution and time resolution that is required in the analysis of nonstationary signals. We saw this before in Example 10.9. If the window is too long, the signal properties may change too much across the window; if the window is too short, resolution of narrowband components will be sacrificed. This trade-off is illustrated in the following example.

Example 10.11 Spectrogram Display of the Time-Dependent Fourier Transform of Speech

Figure 10.22(a) shows a spectrogram display of the time-dependent Fourier transform of the speech signal in Figure 10.21. The time waveform is also shown on the same time scale, below the spectrogram. More specifically, Figure 10.22(a) is a *wideband spectrogram*. A wideband spectrogram representation results from a window that is relatively short in time and is characterized by poor resolution in the frequency dimension and good resolution in the time dimension. The frequency axis is labeled in terms of continuous-time frequency. Since the sampling rate of the signal was 16,000 samples/s, it follows that the frequency $\lambda = \pi$ corresponds to 8 kHz. The specific window used in Figure 10.22(a) was a Hamming window of duration 6.7 ms, corresponding to $L = 108$. The value of R was 16, representing 1-ms time increments.⁷ The broad, dark bars that move horizontally across the spectrogram correspond to the resonance frequencies of the vocal tract, which, as we see, change with time. The vertically striated appearance of the spectrogram is due to the quasiperiodic nature of voiced portions of the waveform, as is evident by comparing the variations in the waveform display and the spectrogram. Since the length of the analysis window is on the order of the length of a period of the waveform, as the window slides along in time, it alternately covers high-energy segments of the waveform and then lower energy segments in between, thereby producing the vertical striations in the plot during voiced intervals.

In a *narrowband* time-dependent Fourier analysis, a longer window is used to provide higher frequency resolution, with a corresponding decrease in time resolution.

⁷In plotting spectrograms, it is common to use relatively small values of R so that a smoothly varying display is obtained.

Such a narrowband analysis of speech is illustrated by the display in Figure 10.22(b). In this case, the window was a Hamming window of duration 45 ms. This corresponds to $L = 720$. The value of R was again 16.

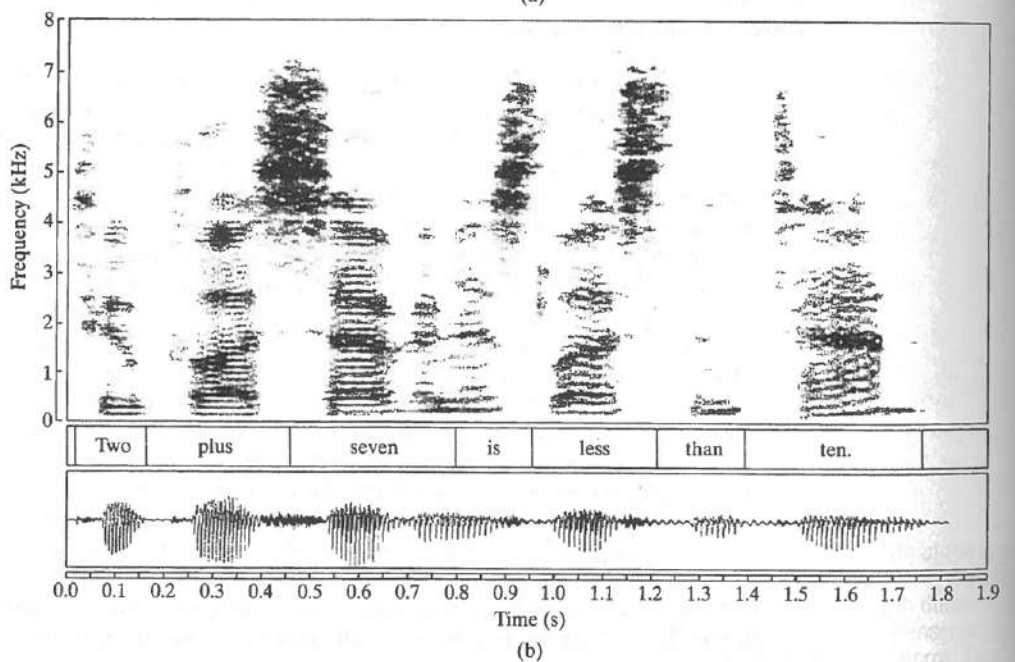
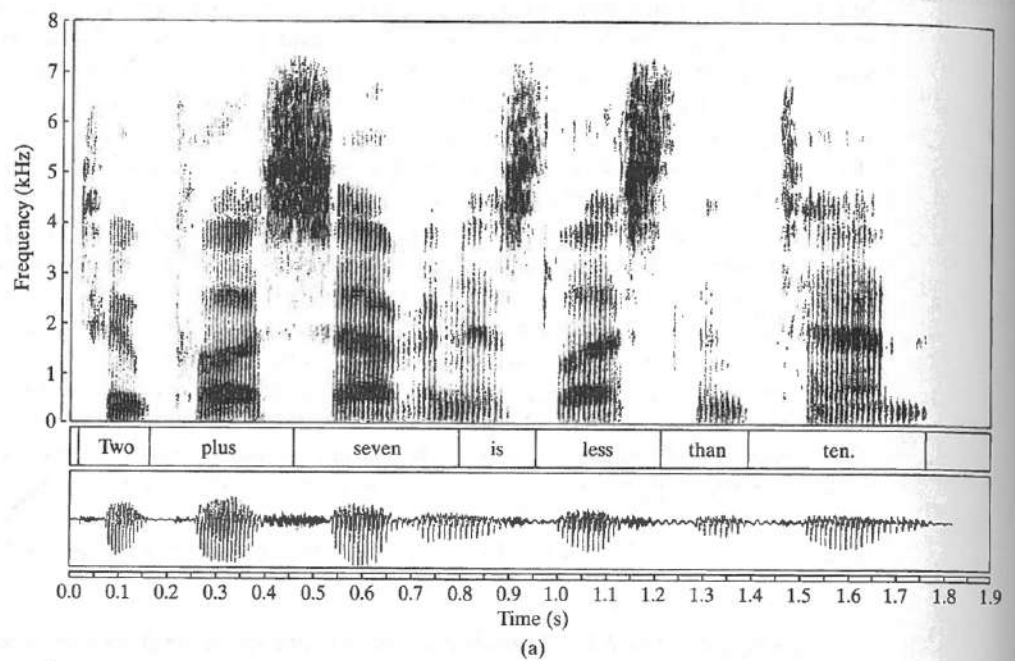


Figure 10.22 (a) Wideband spectrogram of waveform of Figure 10.21. (b) Narrowband spectrogram.

This example only hints at the many reasons that the time-dependent Fourier transform is so important in speech analysis and processing. Indeed, the concept is used directly and indirectly as the basis for acoustic-phonetic analysis and for many fundamental speech-processing applications, such as digital coding, noise and reverberation removal, speech recognition, speaker verification, and speaker identification. For present purposes, our discussion simply serves as an introductory illustration.

10.4.2 Time-Dependent Fourier Analysis of Radar Signals

Another application area in which the time-dependent Fourier transform plays an important role is radar signal analysis. The following are elements of a typical radar system based on the Doppler principle:

- Antennas for transmitting and receiving (often the same).
- A transmitter that generates an appropriate signal at microwave frequencies. In our discussion, we will assume that the signal consists of sinusoidal pulses. While this is often the case, other signals may be used, depending on the specific radar objectives and design.
- A receiver that amplifies and detects echoes of the transmitted pulses that have been reflected from objects illuminated by the antenna.

In such a radar system, the transmitted sinusoidal signal propagates at the speed of light, reflects off the object, and returns at the speed of light to the antenna, thereby undergoing a time delay of the round-trip travel time from the antenna to the object. If we assume that the transmitted signal is a sinusoidal pulse of the form $\cos(\Omega_0 t)$ and the distance from the antenna to the object is $\rho(t)$, then the received signal is a pulse of the form

$$s(t) = \cos[\Omega_0(t - 2\rho(t)/c)], \quad (10.58)$$

where c is the velocity of light. If the object is not moving relative to the antenna, then $\rho(t) = \rho_0$, where ρ_0 is the range. Since the time delay between the transmitted and received pulses is $2\rho_0/c$, a measurement of the time delay may be used to estimate the range. If, however, $\rho(t)$ is not constant, the received signal is an angle-modulated sinusoid and the phase difference contains information about both the range and the relative motion of the object with respect to the antenna. Specifically, let us represent the time-varying range in a Taylor's series expansion as

$$\rho(t) = \rho_0 + \dot{\rho}_0 t + \frac{1}{2!} \ddot{\rho}_0 t^2 + \dots, \quad (10.59)$$

where ρ_0 is the nominal range, $\dot{\rho}_0$ is the velocity, $\ddot{\rho}_0$ is the acceleration, and so on. Assuming that the object moves with constant velocity (i.e., $\ddot{\rho}_0 = 0$), and substituting Eq. (10.59) into Eq. (10.58), we obtain

$$s(t) = \cos[(\Omega_0 - 2\Omega_0 \dot{\rho}_0/c)t - 2\Omega_0 \rho_0/c]. \quad (10.60)$$

In this case, the frequency of the received signal differs from the frequency of the transmitted signal by the *Doppler frequency*, defined as

$$\Omega_d = -2\Omega_0 \dot{\rho}_0/c. \quad (10.61)$$

Thus, the time delay can still be used to estimate the range, and we can determine the speed of the object relative to the antenna if we can determine the Doppler frequency.

In a practical setting, the received signal is generally very weak, and thus a noise term should be added to Eq. (10.60). We will neglect the effects of noise in the simple analysis of this section. Also, in most radar systems, the signal of Eq. (10.60) would be frequency shifted to a lower nominal frequency in the detection process. However, the Doppler shift will still satisfy Eq. (10.61), even if $s(t)$ is demodulated to a lower center frequency.

To apply time-dependent Fourier analysis to such signals, we first bandlimit the signal to a frequency band that includes the expected Doppler frequency shifts and then sample the resulting signal with an appropriate sampling period T , thereby obtaining a discrete-time signal of the form

$$x[n] = \cos[(\omega_0 - 2\omega_0 \dot{\rho}_0/c)n - 2\omega_0 \rho_0/c], \quad (10.62)$$

where $\omega_0 = \Omega_0 T$. In many cases, the object's motion would be more complicated than we have assumed, requiring the incorporation of higher order terms from Eq. (10.59) and thereby producing a more complicated angle modulation in the received signal. Another way to represent this more complicated variation of the frequency of the echoes is to use the time-dependent Fourier transform with a window that is short enough, so that the assumption of constant Doppler-shifted frequency is valid across the entire window interval, but not so short as to sacrifice adequate resolution when two or more moving objects create Doppler-shifted return signals that are superimposed at the receiver.

Example 10.12 Time-Dependent Fourier Analysis of Doppler Radar Signals

An example of time-dependent Fourier analysis of Doppler radar signals is shown in Figure 10.23. (See Schaefer, Schafer and Mersereau, 1979.) The radar data had been preprocessed to remove low-velocity Doppler shifts, leaving the variations displayed in the figure. The window for the time-dependent Fourier transform was a Kaiser window with $N = L = 64$ and $\beta = 4$. In the figure, $|X_T[k]|$ is plotted with time as the vertical dimension (increasing upward) and frequency as the horizontal dimension.⁸ In this case, the successive DFTs are plotted close together. A hidden-line elimination algorithm is used to create a two-dimensional view of the time-dependent Fourier transform. To the left of the center line is a strong peak that moves in a smooth path through the time-frequency plane. This corresponds to a moving object whose velocity varies in a regular manner. The other broad peaks in the time-dependent Fourier transform are due to noise and spurious returns called *clutter* in radar terminology. An example of motion that might create such a variation of the Doppler frequency is a rocket moving at constant velocity but rotating about its longitudinal axis. A peak moving through the time-dependent Fourier transform might correspond to reflections from a fin on the rocket that is alternately moving toward and then away from the antenna because of the spinning of the rocket. Figure 10.23(b) shows an estimate of the Doppler frequency as a function of time. This estimate was obtained simply by locating the highest peak in each DFT.

⁸The plot shows the negative frequencies on the left of the line through the center of the plot and positive frequencies on the right. This can be achieved by computing the DFT of $(-1)^n x_T[n]$ and noting that the computation effectively shifts the origin of the DFT index to $k = N/2$. Alternatively, the DFT of $x_T[n]$ can be computed and then reindexed.

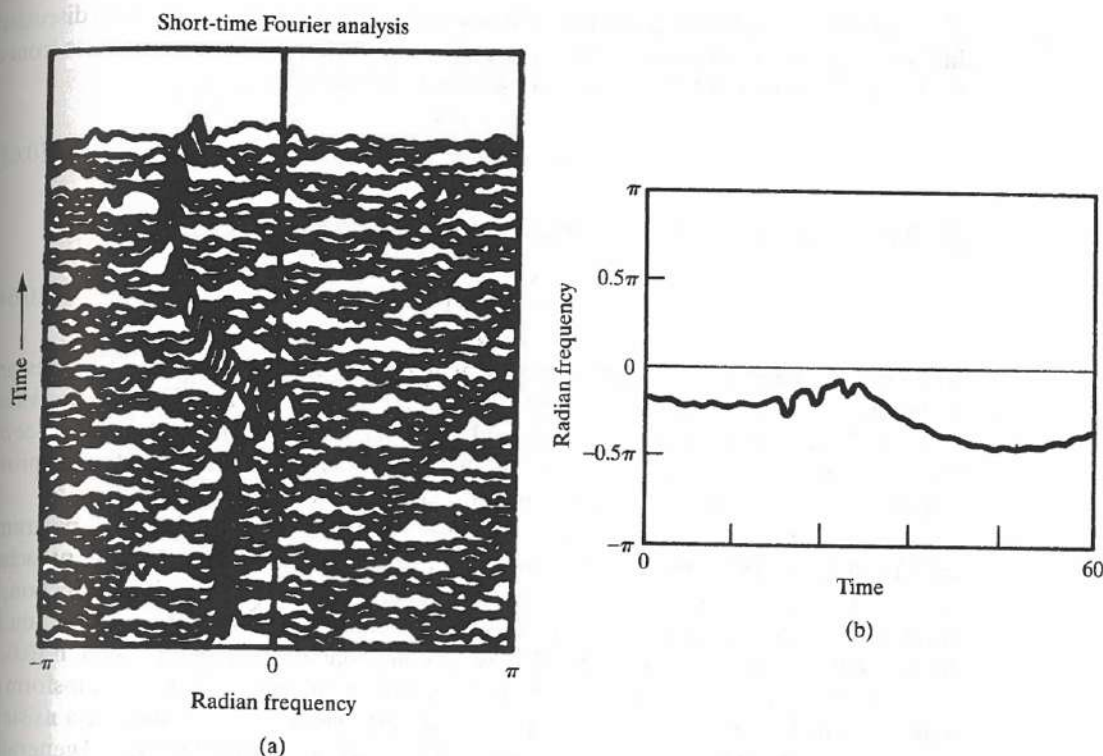


Figure 10.23 Illustration of time-dependent Fourier analysis of Doppler radar signal. (a) Sequence of Fourier transforms of Doppler radar signal. (b) Doppler frequency estimated by picking the largest peak in the time-dependent Fourier transform.

10.5 FOURIER ANALYSIS OF STATIONARY RANDOM SIGNALS: THE PERIODOGRAM

In previous sections, we discussed and illustrated Fourier analysis for sinusoidal signals with stationary (nontime-varying) parameters and for nonstationary signals such as speech and radar. In cases where the signal can be modeled by a sum of sinusoids or a linear system excited by a periodic pulse train, the Fourier transforms of finite-length segments of the signal have a convenient and natural interpretation in terms of Fourier transforms, windowing, and linear system theory. However, more noise-like signals, such as the example of unvoiced speech in Section 10.4.1, are best modeled as random signals.

As we discussed in Section 2.10 and as shown in Appendix A, random processes are often used to model signals when the process that generates the signal is too complex for a reasonable deterministic model. Typically, when the input to an LTI system is modeled as a stationary random process, many of the essential characteristics of the input and output are adequately represented by averages, such as the mean value (dc level), variance (average power), autocorrelation function, or power density spectrum.